=== UNIT 4: CELL CYCLE CONTROL, MITOSIS, MEIOSIS & APOPTOSIS ===

>> 4.1 Eukaryotic Cell Cycle Phases and Checkpoint Control

The eukaryotic cell division cycle is an ordered series of biochemical events culminating in cellular duplication. It is partitioned into Interphase (G1, S, G2) and Mitosis (M phase), governed by an evolutionarily conserved oscillatory biochemical network.

* Cell Cycle Phases: G1 (Gap 1: cell growth, biosyntheses of dNTPs and enzymes), S Phase (Synthesis: replication of genomic DNA and duplication of centrosomes), G2 (Gap 2: proofreading, synthesis of mitotic cyclins and tubulin), and M Phase (karyokinesis and cytokinesis). Non-dividing quiescent cells exit into G0.
* The Cyclin-CDK Engine: Cyclin-Dependent Kinases (CDKs 1, 2, 4, 6) are serine/threonine kinases whose catalytic activity requires binding of regulatory Cyclin subunits (A, B, D, E). CDK levels remain constant throughout the cycle, whereas Cyclin concentrations fluctuate sharply via transcriptional regulation and targeted ubiquitin-proteasome degradation.
* Sequential Cyclin-CDK Complexes in Mammalian Cells:
- G1 phase: Cyclin D binds CDK4 and CDK6 (initiates Rb phosphorylation).
- G1/S transition: Cyclin E binds CDK2 (completes Rb hyperphosphorylation, firing replication origins).
- S phase: Cyclin A binds CDK2 (promotes DNA replication and blocks re-replication).
- G2/M transition: Cyclin A and Cyclin B bind CDK1 / Cdc2 (Maturation Promoting Factor, MPF).
* CDK Regulation: Phosphorylation at Thr161 by CAK (CDK-Activating Kinase) activates the enzyme. Inhibitory phosphorylation at Thr14 and Tyr15 by Wee1 kinase holds the Cyclin B-CDK1 complex inactive during G2. At the onset of mitosis, the dual-specificity phosphatase Cdc25 dephosphorylates Thr14/Tyr15, unleashing active MPF.
* CDK Inhibitors (CKIs): Two major families: INK4 family (p16, p15, p18, p19; specifically inhibit CDK4/6) and Cip/Kip family (p21, p27, p57; broadly inhibit Cyclin E-CDK2, Cyclin A-CDK2, and Cyclin B-CDK1).

>> 4.2 Molecular Checkpoint Cascades

* G1/S Restriction Point (R-Point): Prior to the R-point, cells require continuous mitogenic growth factors. Hypophosphorylated Retinoblastoma protein (pRb) binds and sequesters E2F transcription factors. Mitogen signaling activates Cyclin D-CDK4/6, which partially phosphorylates pRb. Partially liberated E2F transcribes Cyclin E. Cyclin E-CDK2 hyperphosphorylates pRb in a positive feedback loop, releasing free E2F to transcribe DNA polymerase, PCNA, thymidine kinase, and Cyclin A, committing the cell irreversibly to DNA replication.

* DNA Damage Checkpoint (ATM/ATR - p53 Axis): Double-strand breaks (DSBs) activate ATM kinase; single-strand breaks/stalled forks activate ATR kinase. ATM/ATR phosphorylate and activate Chk2/Chk1 checkpoint kinases. Chk2 phosphorylates p53 at Ser20, disrupting its association with Mdm2 (an E3 ubiquitin ligase that normally targets p53 for proteasomal destruction). Stabilized, accumulated p53 functions as a homotetrameric transcription factor driving expression of p21 (Cip1), GADD45 (DNA repair), and PUMA/Bax (apoptosis if damage is irreparable). p21 halts Cyclin E-CDK2, arresting the cycle in G1.
* Spindle Assembly Checkpoint (SAC): Prevents premature anaphase until all kinetochores are attached amphitelically to mitotic spindle microtubules under bipolar tension. Unattached kinetochores assemble the Mitotic Checkpoint Complex (MCC: Mad2, BubR1, Bub3, Cdc20). MCC inhibits the Anaphase-Promoting Complex/Cyclosome (APC/C). Once all kinetochores are aligned at the metaphase plate, SAC turns off, releasing Cdc20 to activate APC/C.

>> 4.3 Mitosis vs. Meiosis and Chromosome Dynamics

* Mitosis: Equational division generating two genetically identical diploid daughters. Stages: Prophase (chromatin condensation, centrosome separation), Prometaphase (nuclear envelope breakdown, kinetochore capture), Metaphase (bipolar equatorial alignment), Anaphase (APC/C degrades Securin, releasing Separase to cleave Cohesin Scc1/Rad21 ring, allowing poleward sister chromatid separation), Telophase (nuclear envelope reassembly, decondensation), Cytokinesis (cleavage furrow formed by actin-myosin contractile ring).
* Meiosis: Reductional division producing 4 genetically diverse haploid gametes. Prophase I comprises 5 substages: Leptotene (chromatin condensation), Zygotene (synaptonemal complex assembly), Pachytene (homologous recombination / crossing over mediated by Rad51/Dmc1), Diplotene (synaptonemal complex dissolves, chiasmata visible), Diakinesis (terminalization of chiasmata). Meiosis I separates homologous chromosome pairs (reductional); Meiosis II separates sister chromatids (equational).

>> 4.4 Apoptosis: Programmed Cell Death Pathways

* Morphological Hallmarks: Cell shrinkage, membrane blebbing, chromatin pyknosis (condensation), karyorrhexis (nuclear fragmentation), internucleosomal DNA fragmentation into 180-200 bp ladders by Caspase-Activated DNase (CAD), and phosphatidylserine externalization without inflammatory response.
* Caspases: Cysteine-dependent aspartate-directed proteases. Synthesized as inactive zymogens. Initiator caspases (caspase-8, -9, -10) possess death effector domains (DED) or caspase recruitment domains (CARD) and undergo proximity-induced autoactivation. Executioner caspases (caspase-3, -6, -7) cleave cellular substrates (ICAD, PARP, lamin, alpha-fodrin).
* Extrinsic Pathway: FasL or TNF-alpha binds trimeric death receptors (Fas/CD95, TNFR1). The intracellular Death Domain (DD) recruits FADD adaptor, which recruits

procaspase-8 via DED to form the Death-Inducing Signaling Complex (DISC). Dimerized caspase-8 activates and directly cleaves executioner caspase-3.
* Intrinsic (Mitochondrial) Pathway: Cellular stress (DNA damage, oxidative stress, oncogene activation) triggers BH3-only pro-apoptotic proteins (Bim, Bid, Puma, Noxa) to neutralize anti-apoptotic Bcl-2/Bcl-xL and directly activate Bax and Bak. Bax/Bak oligomerize in the OMM, creating Mitochondrial Outer Membrane Permeabilization (MOMP). Cytochrome c leaks into the cytosol and binds Apaf-1 in the presence of dATP, assembling a heptameric wheel-like 'Apoptosome'. The apoptosome recruits and activates procaspase-9, which activates executioner caspase-3.

[EXAMPLE 1: Retinoblastoma (Rb) Phosphorylation Kinetics and the R-Point]

Problem Context: Explain how mutations in the CDKN2A gene (encoding p16-INK4a) or amplification of CCND1 (Cyclin D1) drive oncogenesis by dysregulating the G1/S restriction point.
Normal Molecular Circuitry:
1. Quiescent cells in G0 maintain Rb in an unphosphorylated state bound to transactivation domains of E2F1-3, recruiting HDACs to silence E2F-responsive promoters.
2. p16-INK4a binds CDK4/6, blocking its association with Cyclin D.
Oncogenic Dysregulation Case 1 (Loss of p16-INK4a):
- Deletion or promoter hypermethylation of CDKN2A removes all brakes on CDK4/6.
- Cyclin D-CDK4/6 is constitutively active, driving continuous phosphorylation of Rb at Ser780 and Ser795.
Oncogenic Dysregulation Case 2 (Cyclin D1 Amplification):
- Common in mantle cell lymphomas and breast carcinomas.
- Abundant Cyclin D titrates out basal levels of CKIs, driving relentless partial phosphorylation of Rb.
Downstream Consequences: Partially phosphorylated Rb releases repression; E2F transcribes Cyclin E. Cyclin E-CDK2 hyperphosphorylates Rb at 16 distinct Ser/Thr residues, fully liberating E2F. Cells bypass the requirement for growth factor mitogens, acquire self-sufficiency in growth signals, and undergo uncontrolled proliferation.
Conclusion: Pharmaceutical CDK4/6 inhibitors (Palbociclib, Ribociclib) bind the ATP-binding pocket of CDK4/6, restoring the unphosphorylated tumor-suppressive state of Rb.

[EXAMPLE 2: Apoptosome Heptamer Assembly Stoichiometry and Caspase Activation]

Problem Context: Calculate the molecular weight and subunit stoichiometry of the human Apoptosome platform, and explain why dATP/ATP hydrolysis is mandatory for caspase-9 activation.
Subunit Components:
- Apaf-1 (Apoptotic Protease Activating Factor-1): 140 kDa monomer consisting of

an N-terminal CARD domain, a nucleotide-binding oligomerization domain (NOD), and two C-terminal WD40 beta-propeller domains.
- Cytochrome c: 12 kDa heme protein released from mitochondrial IMS.
- Procaspase-9: 46 kDa zymogen containing an N-terminal CARD domain and catalytic large (p20) and small (p10) subunits.
Assembly Cascade:
1. In healthy cells, Apaf-1 exists in an auto-inhibited monomeric state where WD40 domains fold back over the NOD domain.
2. Upon MOMP, released cytochrome c binds specifically to the WD40 repeats.
3. Binding drives exchange of dADP for dATP/ATP, triggering a dramatic conformational opening.
4. Seven opened Apaf-1:cytochrome c complexes oligomerize radially via their NOD domains into a 1.4 MDa wheel-shaped heptamer.
5. The central hub clusters seven CARD domains into an elevated disk called the CARD ring.
6. Procaspase-9 molecules bind the Apaf-1 CARD ring via CARD-CARD homophilic interactions. Forced proximity induces dimerization of procaspase-9 monomers, creating the active catalytic site.
Conclusion: Unlike executioner caspases which require proteolytic cleavage, initiator caspase-9 is activated purely by allosteric dimerization on the multimeric apoptosome scaffolding platform.

[EXAMPLE 3: Analysis of Internucleosomal DNA Laddering vs. Necrotic Smear]

Problem Context: In a biotechnology forensic toxicology experiment, cells are treated with either Staurosporine (an apoptosis inducer) or Hydrogen Peroxide (causing acute necrosis). Explain why agarose gel electrophoresis produces a discrete 180-200 bp ladder in apoptotic cells, but a diffuse smear in necrotic cells.
Apoptotic Internucleosomal Cleavage:
- Active caspase-3 cleaves the inhibitory chaperone ICAD (Inhibitor of Caspase-Activated DNase).
- Liberated catalytic CAD endonuclease translocates to the nucleus.
- CAD specifically targets naked linker DNA regions (~20-50 bp) between adjacent nucleosome core particles, leaving histone-wrapped 147 bp core DNA protected.
- Cleavage produces discrete DNA fragments that are integer multiples of the nucleosomal unit (~180-200 bp, 360-400 bp, 540-600 bp, 720-800 bp).
- On a 1.5% agarose gel, this displays as a sharp, highly reproducible ladder pattern.
Necrotic Non-Specific Degradation:
- Necrosis involves early plasma membrane rupture and influx of extracellular Ca^{2+} and water.
- Lysosomal membranes rupture hydrolytically, dumping non-specific endo- and exo-nucleases (DNase I, DNase II) into the chromatin without regulation.
- DNA is digested randomly at all sites, yielding a continuous distribution of

fragment sizes showing as an indistinct continuous smear on agarose gels.
Conclusion: Internucleosomal DNA laddering is a golden diagnostic hallmark distinguishing physiological programmed cell death from uncontrolled accidental necrosis.

[EXAMPLE 4: Flow Cytometric Propidium Iodide Cell Cycle Analysis]

Problem Context: Describe how DNA content analysis using Propidium Iodide (PI) fluorescence flow cytometry quantifies the percentage of cells in G0/G1, S, G2/M phases, and sub-G1 apoptotic fractions.
Principle: Propidium Iodide intercalates stoichiometrically between DNA base pairs (1 molecule per 4-5 bp) without sequence bias. Fluorescence emission intensity (FL2 at 617 nm) is directly proportional to cellular DNA content.
Histogram Profiles:
1. G0/G1 Phase Peak: Diploid cells have 2N DNA content. Appears as a tall, sharp peak at arbitrary fluorescence channel ~200.
2. G2/M Phase Peak: Cells have duplicated their genome (4N DNA content). Appears as a distinct peak at exactly twice the fluorescence channel of G1 (~400).
3. S Phase Distribution: Intermediate fluorescence continuum (between 200 and 400 channels) representing actively replicating cells with DNA content between 2N and 4N.
4. Sub-G1 (Apoptotic) Peak: Apoptotic cells lose low-molecular-weight internucleosomal DNA fragments during washing/permeabilization. They exhibit hypodiploid DNA content (<2N), appearing as a distinct peak below channel 200.
Mathematical Quantification: Deconvolution algorithms (Dean-Jett-Fox or Watson Pragmatic models) fit Gaussian curves to quantify the precise mitotic fraction and evaluate drug-induced cell cycle arrests.

>> 4.5 University Review & Examination Questions

* Q1: Explain how the Anaphase-Promoting Complex/Cyclosome (APC/C) coordinates the metaphase-to-anaphase transition and mitotic exit.
Answer: APC/C is a multi-subunit E3 ubiquitin ligase. At metaphase, once the spindle assembly checkpoint is satisfied, Cdc20 activates APC/C. APC/C-Cdc20 targets Securin for polyubiquitination and 26S proteasomal degradation. Degradation of Securin releases active Separase (a protease). Separase cleaves the Scc1/Rad21 subunit of the Cohesin ring complex holding sister chromatids together, allowing spindle microtubules to pull chromatids to opposite poles. During late anaphase and telophase, APC/C switches co-activators from Cdc20 to Cdh1. APC/C-Cdh1 targets Cyclin B and Aurora/Polo kinases for destruction, extinguishing CDK1 activity and permitting spindle disassembly, nuclear envelope reformation, and cytokinesis.
* Q2: Compare the extrinsic and intrinsic pathways of apoptosis with respect to triggers, initiator caspases, adaptor proteins, and cross-talk mechanisms.
Answer: Extrinsic pathway is triggered by extracellular ligand binding to death

receptors (FasL/Fas, TNF/TNFR1), uses adaptor FADD to recruit procaspase-8 forming the DISC. Intrinsic pathway is triggered by intracellular genotoxic stress, uses BH3-only proteins to trigger Bax/Bak-mediated MOMP, releasing cytochrome c to form the Apaf-1 apoptosome which activates procaspase-9. Cross-talk occurs via Caspase-8-mediated proteolytic cleavage of the BH3-only protein Bid into truncated Bid (tBid). tBid translocates to mitochondria to activate Bax/Bak, amplifying the death signal through the intrinsic pathway.

>> 4.6 Key Takeaways and Revision Points

* Cyclin-CDK oscillation drives cell cycle transitions; Cyclin D-CDK4/6 and Cyclin E-CDK2 govern the G1/S Restriction Point.
* p53 is stabilized via ATM/Chk2 upon DNA damage, transcribing p21 (Cip1) to induce G1 arrest or Puma/Bax to trigger apoptosis.
* Apoptosis proceeds through extrinsic (Death receptor/Caspase-8) or intrinsic (Mitochondria/Apaf-1/Caspase-9) pathways converging on Executioner Caspase-3.
* CAD endonuclease executes internucleosomal DNA fragmentation (180-200 bp ladders), while Bcl-2 proteins govern mitochondrial membrane integrity.